Data Science for Political Campaigns

Week 8 — Can we trust polls?

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October 28, 2026

Roadmap

- **Last week:** experiments can mislead when you condition on compliance. The vendor's $+5.5$ was selection bias, not an ad effect.
- **This week:** three polls of the same race disagree by 12 points. Why, and what to do about it.
- **Next week:** cost-effectiveness across tactics.

This Week's Case

Case: You are the campaign manager for a competitive state Senate race

Eight weeks before Election Day. Three polls of your race:

Poll	Method	Surveyed	n	Result
Landmark Research	Phone	Likely voters	$\sim$1,000	Rep +4
Progressive Data	Online	Likely voters	$\sim$1,000	Dem +8
State University	Mixed	Registered voters	$\sim$1,200	Dem +1

You need to allocate \$2M. Which poll do you trust?

Discuss

From this week's reading

Four this week. Cohn, "We Gave Four Good Pollsters the Same Raw Data" (five results off *one* dataset) and Cohn, "The Big Changes Coming to the Times/Siena Poll." Pew, "How different weighting methods work," and Pew's *Polling Basics*, Lessons 4 and 6.

From the reading: one reason two good pollsters can publish different numbers off the same raw data, and whether either did anything wrong.

- The phone poll and online poll disagree by 12 points. Why?
- What kind of voter answers a phone call from an unknown number? What kind signs up for an online panel?
- If the phone poll has too many old voters and old voters lean Republican, what happens when you reweight?
- After 2016 and 2020, should we just add a few points to the Republican every time?

Pair up. Three minutes.

Why Polls Disagree

Collecting the data

- **Sampling error.** Random noise from interviewing 1,000 people instead of 500,000.
- **Nonresponse bias.** The people who respond differ systematically from the people who don't.
- **Mode effects.** The same person answers a live interviewer differently than an online survey.

Analyzing the data

- **Weighting.** Which groups the pollster adjusts, and to what targets.
- **Likely-voter screen.** Which respondents count as voters.

The Margin of Error Is a Confidence Interval

A poll asks about 1,000 people to estimate what *all* voters think. Ask a different 1,000 and you get a different number.

The margin of error measures that noise. It is the Week 5 confidence interval, on a poll number instead of a treatment effect.

At $n \approx 1{,}000$:

$$\text{margin of error} \;=\; 1.96 \times \underbrace{1.6}_{\text{standard error}} \;\approx\; \pm 3 \text{ points}$$

"Dem 48%" means "Dem somewhere around 45–51%."

Standard Error

$$\text{standard error} = \frac{\text{how far a typical person is from the average}}{\sqrt{n}}$$

- The **bottom** is sample size. Week 5: four times the sample, half the SE.
- The **top** is how spread out people are on the thing we are measuring.

For a yes/no question, the top is $\sqrt{p(1-p)}$, where p is the share who say yes. That is Week 5's $p(1-p)$, square-rooted.

Our phone poll: 46% of 1,001 respondents picked the Democrat.

$$\frac{\sqrt{0.46 \times 0.54}}{\sqrt{1{,}001}} = \frac{0.50}{31.6} = 0.016 = 1.6 \text{ points}$$

What Is a Lead?

Lead: one candidate's share minus the other's. Every number in this week's case is a lead: Rep $+4$, Dem $+8$, Dem $+1$.

One respondent switches from the Democrat to the Republican. The Democrat's share falls by one person *and* the Republican's rises by one, so the lead moves by two.

So a lead bounces around more than either candidate's share, and it gets its own standard error.

Standard Error of a Lead

Score each respondent $+1$ for the Democrat, -1 for the Republican, 0 for undecided. The average score is the lead.

	Answers	**Typical distance from average**	**Standard error**	**Margin of error**
"Picked the Democrat?"	0 or 1	about 0.5	1.6	± 3
"Which way did they lean?"	-1, 0, $+1$	about 1.0	3.1	± 6

Why about 1.0? Almost everyone picks a side, so nearly every score is $+1$ or -1 and the average sits near 0. A typical respondent is about 1 away from it.

Same n, about twice the variation, about twice the standard error.

Is It a Real Lead, or a Statistical Tie?

Our three polls, against the ± 6 margin of error on a lead:

- **University, Dem +1:** inside the margin. A statistical tie.
- **Phone, Rep +4:** also inside ± 6. A statistical tie: anything from R+10 to D+2.
- **Online, Dem +8:** clears ± 6, but is consistent with a 2-point squeaker.

This is the Week-5 significance test: is the lead distinguishable from zero?

Is the 12-Point Gap Even Real?

Two of our polls disagree by 12 points. Is that difference distinguishable from noise?

Each poll's lead has a standard error of 3.1 points. The standard error of the difference between them:

$$\sqrt{3.1^2 + 3.1^2} \;=\; 4.4 \text{ points}$$

95% confidence interval:

$$12 \pm 1.96 \times 4.4 \;=\; [+3, +21]$$

The interval excludes zero, so the difference is statistically significant. But it could be anywhere from 3 to 21 points.

The Margin of Error Is One Type of Uncertainty

The margin of error measures only sampling error: the luck of which 1,000 people you include in the poll. It shrinks as n grows. It says nothing about who responds.

Bias does not shrink when you collect more of the same kind of data. A poll of 50,000 biased respondents is a more precise wrong answer.

A margin of error on a lead says *this poll might be off by six points in either direction, at random*. In 2020, the polls were off by about four points on the lead, nearly all in the same direction.

You can *compute* the margin of error. You cannot compute the bias, only argue about its direction.

Nonresponse Bias

Nonresponse bias: when the people who respond to the survey differ systematically from the people who don't, in ways that affect the estimate.

Plain-English version: if Republican-leaning voters are less likely to pick up the phone, the poll will overestimate Democratic support, even if the pollster called the right mix of phone numbers.

Phone response rates are now below **5%**, so the people who do respond are not a random sample. Who responds also depends on how you reach them:

- **Phone polls** skew older. Younger voters don't pick up unknown numbers.
- **Online panels** skew younger and more politically engaged.

Mode Effects

Mode: the method of contact: live phone call, automated call, online panel, text-to-web.

Mode effect: the same person gives a different answer depending on the mode.

Plain-English version: people might say what they think sounds good when a live person is asking (social desirability bias). They do it less on a screen.

Example: the "shy Trump voter." If Trump voters gave live interviewers a different answer, live-phone polls should have missed by more than online polls in 2020. They did not. AAPOR found that every mode overstated the Democrat, with only minor differences between them.

Post-Stratification: A Fix

Post-stratification: adjusting the sample after the fact so that it matches the population on known demographics (age, race, education, party registration).

Plain-English version: if your sample has too many old people, give each young respondent more weight when computing the results.

1. For each demographic group, compute a weight:

$$w_g = \frac{\text{population share of group } g}{\text{sample share of group } g}$$

2. Recompute the results using these weights.

Working the Weights: Our Phone Poll

The phone poll interviewed 412 voters over 65 and 60 voters under 30, in a district that is 25% over 65 and 18% under 30.

Age	In the sample	In the district	Weight	Unweighted Dem share
18–29	6.0%	18.0%	3.00	58.3%
30–44	17.0%	25.0%	1.47	50.6%
45–64	35.9%	32.1%	0.90	46.2%
65+	41.2%	24.9%	0.60	42.5%

Weight = district share ÷ sample share. Each of those 60 young respondents now counts as three people. Each voter over 65 counts as six-tenths of one.

Weighting Example

Unweighted. Each group counts for the share the poll happened to reach:

$$0.060(58.3) + 0.170(50.6) + 0.359(46.2) + 0.412(42.5) = 46.2\%$$

Weighted. Multiply each group by its weight:

$$0.060(3.00)(58.3) + 0.170(1.47)(50.6) + 0.359(0.90)(46.2) + 0.412(0.60)(42.5) = 48.6\%$$

Sample share $\times$ weight = district share: $0.060 \times 3.00 = 0.18$. Weighting swaps the poll's age mix for the district's.

Post-Stratification: The Limits

Post-stratification fixes imbalances on the variables you weight by. It does not fix imbalances *within* those groups.

Example: a fair poll of 1,000 in our district would have about 180 young voters. Ours has 60. Weighting counts each of them three times. It does not find the 120 who are missing.

If the 60 who picked up are the politically engaged young voters, weighting makes the engaged three times louder, and the disengaged are still missing.

Why Not Weight on Everything?

If weighting on age helps, why not weight on everything? Three reasons:

1. **You need a population target.** The Census and the voter file give you age, race, education, party registration. There is no census of "trusts institutions."
2. **Cells get small.** Split our 60 young respondents by education: 30 apiece. Add four regions: about seven.
3. **Small cells make extreme weights.** One respondent can carry a huge weight and move the result.

Pew compared nine online samples. The least biased also weighted on political variables, like party ID and ideology. Pew then tested it directly: which variables you choose matters more than the method.

What Weighting Costs You

Weighting makes the estimate noisier.

Effective sample size: the size of an *unweighted* poll that would be just as precise.

$$n_{\text{eff}} = \frac{(\sum w)^2}{\sum w^2} = \frac{1{,}001^2}{1{,}348} = 743$$

w is each respondent's weight. $\sum$ means add up over all 1,001 respondents.

	Respondents	Margin of error (Dem share)
Unweighted	1,001	±3.1 pp
Weighted by age	743	±3.6 pp

Weighting on age moved the Dem share 2.4 points and cost 258 interviews.

If every weight is 1, $n_{\text{eff}} = n$. The more unequal the weights, the smaller n_{eff}.

One Dataset, Five Answers

Nate Cohn (NYT) gave four pollsters the 867 responses behind one Times/Siena poll of Florida. Each chose how to weight and who counted as a likely voter.

Pollster	**Result**
Charles Franklin (Marquette Law)	Clinton +3
Patrick Ruffini (Echelon Insights)	Clinton +1
Omero, Green & Rosenblatt (Penn Schoen Berland)	Clinton +4
Corbett-Davies, Gelman & Rothschild	Trump +1
The Times's own team	Clinton +1

Polling in 2016

The national polls in 2016 were fine. Clinton +3.2 against a +2.1 result. AAPOR rates them among the most accurate since 1936, at less than half the average error over that span. The state polls in the upper Midwest were not fine.

AAPOR found the most evidence for two causes:

1. Voters who decided in the last week broke for Trump. 11 to 15% of voters in Michigan, Wisconsin, Pennsylvania and Florida chose that late; in Wisconsin they broke for him by 29 points.
2. Many polls did not weight on education, which had just become a very strong predictor of vote choice. Of the state polls in the final fortnight, 18% of Michigan's weighted on it, 27% of Wisconsin's, 33% of Pennsylvania's.

2020 Was Never Explained

National *presidential* polls overstated Biden's margin by 3.9 points, the largest error since 1980. Senate and governor polls overstated the Democrat by even more: 6.0 points.

AAPOR then ruled out, as *primary* sources: late deciders, education weighting (92% of the state polls that disclosed their methods did it this time, against 18–33% in 2016), the demographic mix, shy Trump voters, differential turnout, and the early-vs-Election-Day mix.

That leaves nonresponse. They found it contributed. They could not show it was the main cause: "without knowing how nonrespondents compare to respondents we cannot conclusively identify the primary source of polling error."

We still do not know why the polls missed in 2020.

Likely-Voter Screens

Likely-voter screen: the pollster's method for deciding which respondents to include, based on whether they're likely to actually vote.

Common approaches:

- Self-reported intention: "Do you plan to vote?"
- Past vote history from the voter file
- A composite score combining both

Each one produces a different electorate. Pew ran 16 versions of these methods on the same 2014 survey and got everything from Dem +2 to Rep +7, against a verified-voter benchmark of Rep +3.

From the case: the university poll surveyed registered voters and the other two screened for likely voters.

Does Asking People Work?

Rogers and Aida matched 29,403 pre-election interviews to the voter file and looked up who actually voted.

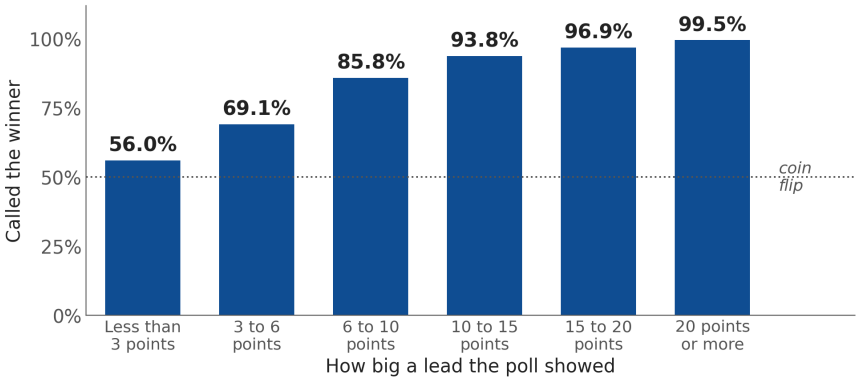
"Will you vote?"	Said it	Actually voted
Yes	9,272	86.7%
Probably	1,004	74.4%
50–50	417	66.7%
No	201	54.7%
Don't know	131	76.3%

About half of those who said they would *not* vote, voted.

The voter file beats asking people: in models that also control for age and gender, past vote history does more than twice the predictive work of the self-reported vote likelihood question.

So Are Polls Any Good?

Everything so far says a poll can be wrong in ways it cannot measure. But do polls work?



Silver Bulletin: polls in the final 21 days before presidential primaries and president, Senate, House and governor general elections since 1998.

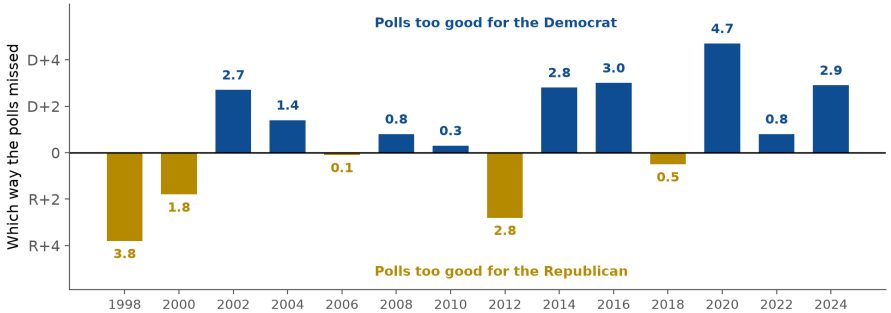
Polls Are Pretty Good

A lead of 15 to 20 points calls the winner 97% of the time. Polls are not broken.

A lead under 3 points calls the winner 56% of the time. That is a coin flip.

Which Way Will They Miss?

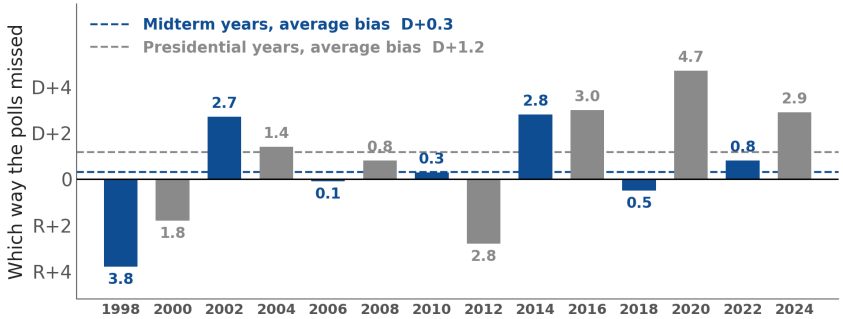
So should you just add a few points to the Republican in every poll?



The last six cycles average D+2.3, and five of the six missed toward the Democrat. But over all fourteen the average is D+0.7, and five of the fourteen missed toward the Republican. Adding points is a bet that the last miss is the next one.

Silver Bulletin, general elections for president, Senate, House and governor since 1998.

Midterms and Presidentials



Same fourteen cycles as the last chart, split by year type.

What Should You Look for in a Poll?

Pew's checklist (*Polling Basics*, Lesson 6), run on our three polls:

	Landmark	**Progressive Data**	**State University**
Who sponsored it?	?	?	?
Who was interviewed?	Likely voters	Likely voters	Registered voters
How was it conducted?	Live phone	Online panel	Phone + online
Random or opt-in?	?	?	?
When was it done?	Past 2 weeks	Past 2 weeks	Past 2 weeks
How many people?	∼1,000	∼1,000	∼1,200
Weighted? How?	?	?	Yes; on what?
Neutral questions?	?	?	?

Every **?** is a question to send the pollster. Pew: if a pollster won't provide this information, "you shouldn't trust its polls."

What Do You Do With Three Polls?

No single poll is "right." Each has known biases.

The campaign manager's job:

1. Understand *which direction* each poll is likely biased.
2. Reweight where possible to correct known imbalances.
3. Acknowledge that unknown biases cannot be fixed by reweighting.
4. Make a decision under uncertainty, rather than trusting the most favorable poll.

Polling is the *least bad* tool for knowing where a race stands. Everything else is worse.

Takeaway

A poll is a sample, not a census. Who responds determines what you see.

- Phone polls skew old. Online polls skew young. Both can be reweighted, partially.
- Post-stratification fixes demographic imbalances but not within-group nonresponse bias.
- Nobody has conclusively explained the 2020 miss. AAPOR ruled out the demographic explanations; nonresponse contributed, but they could not show it was the main cause.
- Polls call a lead of 15–20 points right 97% of the time and a lead under 3 points 56% of the time. They are weakest where campaigns need them.
- No poll is "right." The job is to understand the direction and magnitude of each poll's biases.

Lying With Data This Week

Lying with data: Cherry-picking the poll that tells you what you want to hear. Releasing an internal poll without disclosing methodology. House effects: a pollster's methodology systematically favoring one side.

The Checklist So Far

1. **W1:** Conflating fixed and marginal costs.
2. **W2:** Presenting an observational comparison as a causal effect.
3. **W3:** Applying a result from one setting to a different one.
4. **W4:** Cherry-picking the winning arm from a multi-arm test.
5. **W5:** Treating an underpowered null as evidence of no effect.
6. **W7:** Reporting the complier comparison as a causal effect.
7. **W8:** Cherry-picking polls; house effects; ignoring nonresponse bias.

For Next Week

Read before next week

- Lackey & Singer, "Why the Cost-Per-Vote Formula Needs to Change." Trade press written for Democratic practitioners: read it as an artifact from inside the industry, not a neutral source.
- Three **J-PAL** evaluation summaries: canvassing/phone/mail, leafletting, robocalls. Three tactics, three separate experiments, never compared head-to-head.
- Ord, "The Moral Imperative toward Cost-Effectiveness in Global Health."
- Focus for Democracy, "How We Calculate the Cost of a Vote."

Also due next week: your one-page final project outline. Start it now, not next Tuesday! Happy to talk through a topic: come to office hours.

Break: 5 Minutes

$\sim$ 5 min $\sim$

Live Coding

Open `livecode.ipynb`.

Four parts:

1. Toplines for the phone and online polls
2. Is the 12-point gap just sampling noise?
3. Who responded?
4. Reweight both polls by age

Problem Set This Week

Open `problem_set.ipynb`.

- **Task 1:** Compute unweighted toplines for both polls (pre-filled).
- **Task 1b:** Is the 12-point gap sampling noise? Four short questions off one pre-filled cell.
- **Task 2:** Compare age distributions (pre-filled).
- **Task 3: Write a weighted mean yourself.** Compute the reweighted Dem and Rep shares for the phone poll. This is the one coding task.
- **Task 4: Referee the two pollsters** (250–300 words): the phone and online pollsters each pitch you on why *their* number is the one to trust. Rule on the dispute. Each is right about the other and wrong about itself.